

Burden of disease

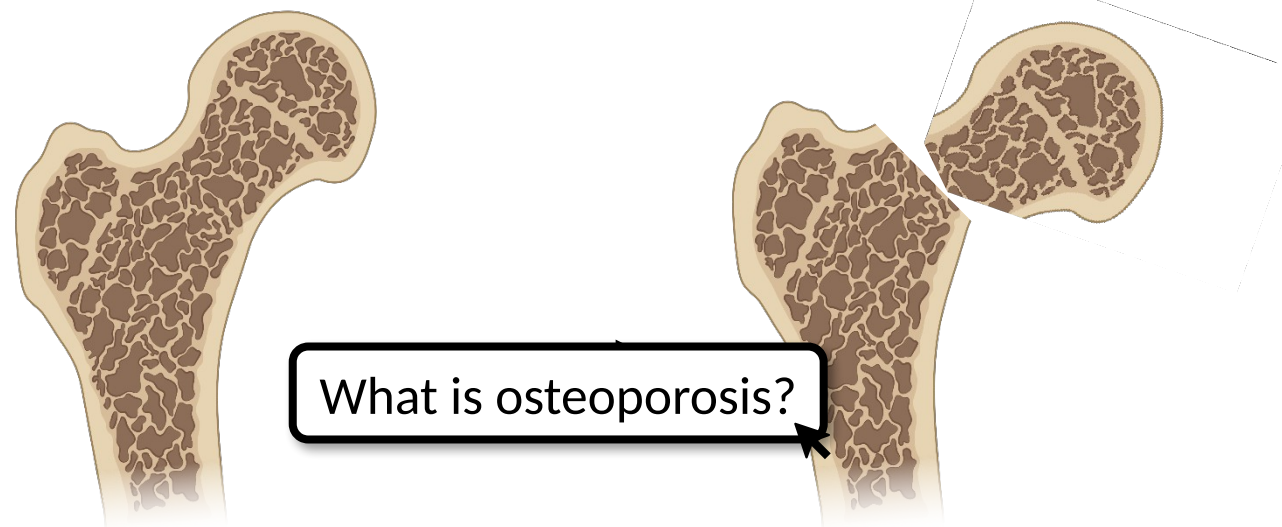
Screening

Diagnostic Criteria

Treatment

Cases

Osteoporosis



What is osteoporosis?

Low bone mass +
Microarchitectural
destruction

Skeletal
fragility

Burden of disease

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Prevalence

What percent of people over the age of 50y have osteoporosis?

~20% of women
~5% of men

Complications

Prevention

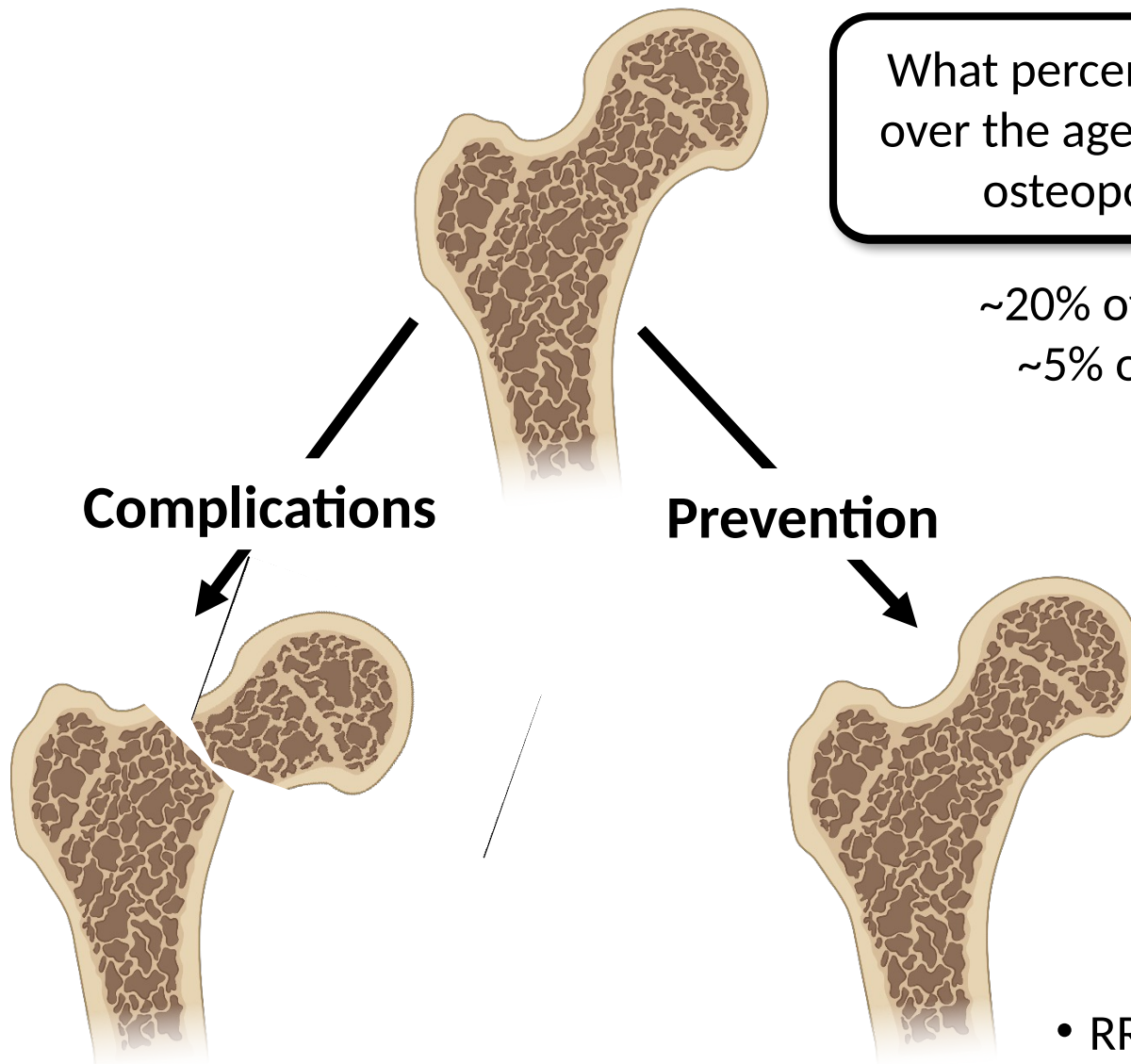
Hip Fracture:

1-year mortality and likelihood of returning to independent living?

- 1 year mortality ~25%
- Return to independence ~50%

RRR and NNT to prevent osteoporotic fracture with bisphosphonates

- RRR: ~ 40% for spin or hip fx
- NNT: ~100 for hip & ~ 15 spine



Burden of disease

Screening

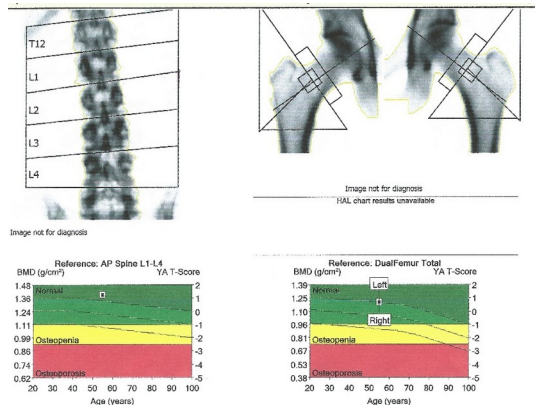
Diagnostic Criteria

Treatment

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Who should be screened?

DXA



1. Women ≥ 65 years old.
2. Postmenopausal women <65 with at least 1 risk factor (consider using **FRAX calculator**)
3. Consider for men with major risk factors.

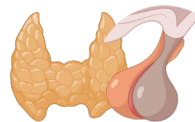
Major risk factors

hint: included in FRAX calculator

Substances



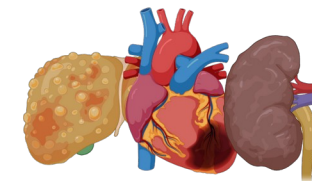
Endocrinopathies



Hyp

g's

Chronic disease/
Malignancy



Medications



Other

Low BMI/ Anorexia
Family hx hip fracture
Previous fracture

Burden of disease

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Diagnostic Criteria

Secondary eval

Treatment

Cases

What are the 3 ways to diagnose osteoporosis?

1

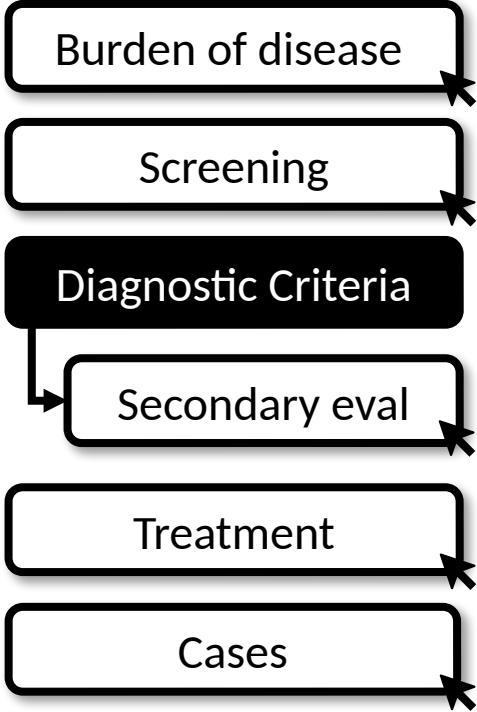
T-score

2

Fragility
Fracture

3

FRAX score



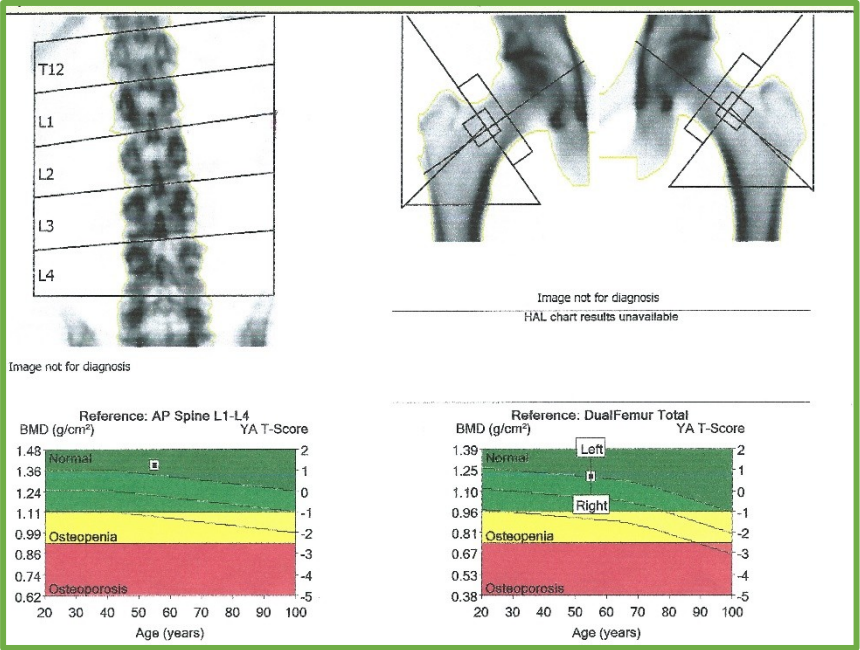
What are the 3 ways to diagnose osteoporosis?

1 T-score

Classification	T-score on DXA
Normal	> -1.0
Osteopenia (LBM)	-1.0 to -2.5
Osteoporosis	< -2.5

2 Fragility Fracture

3 FRAX score



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What are the 3 ways to diagnose osteoporosis?

1

T-score

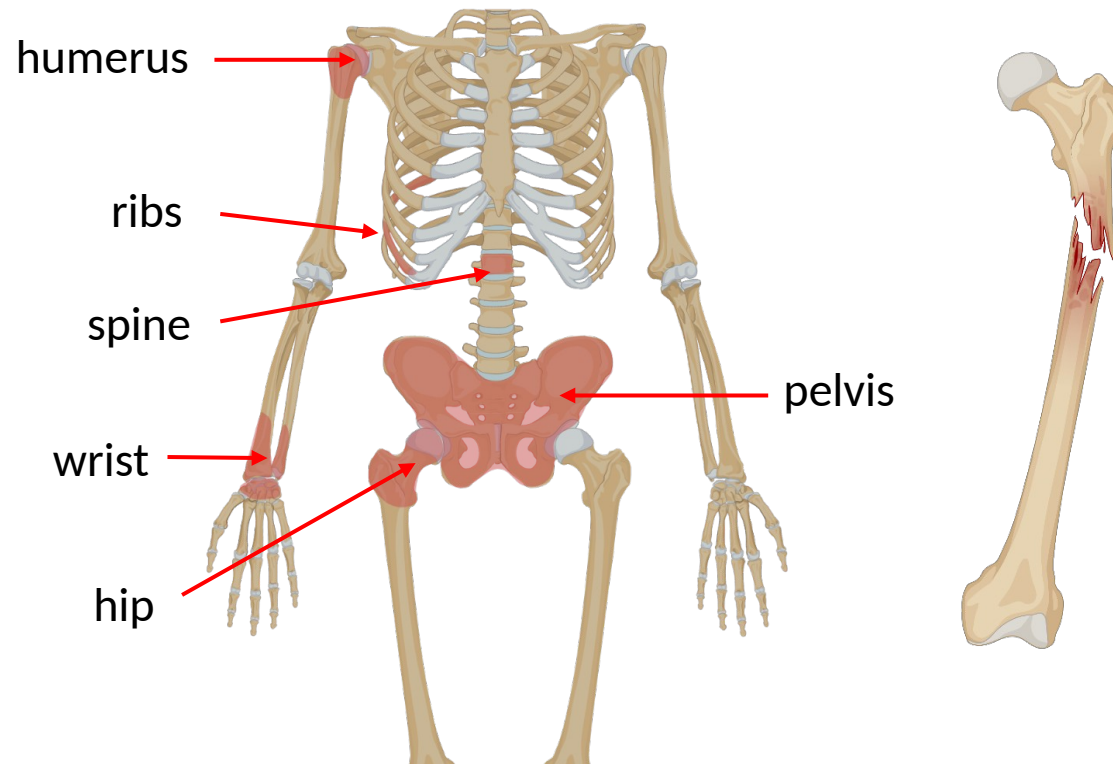
A fracture that occurs spontaneously or with minor trauma (e.g., GLF)

2

Fragility Fracture

3

FRAX score



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What are the 3 ways to diagnose osteoporosis?

1 T-score

2 Fragility Fracture

3 FRAX score

Country: **US (Caucasian)** Name/ID: [About the risk factors](#)

Questionnaire:

1. Age (between 40 and 90 years) or Date of Birth
Age: Y: M: D:

2. Sex ☐ Male ☐ Female

3. Weight (kg)

4. Height (cm)

5. Previous Fracture ☒ No ☐ Yes

6. Parent Fractured Hip ☒ No ☐ Yes

7. Current Smoking ☒ No ☐ Yes

8. Glucocorticoids ☒ No ☐ Yes

9. Rheumatoid arthritis ☒ No ☐ Yes

10. Secondary osteoporosis ☒ No ☐ Yes

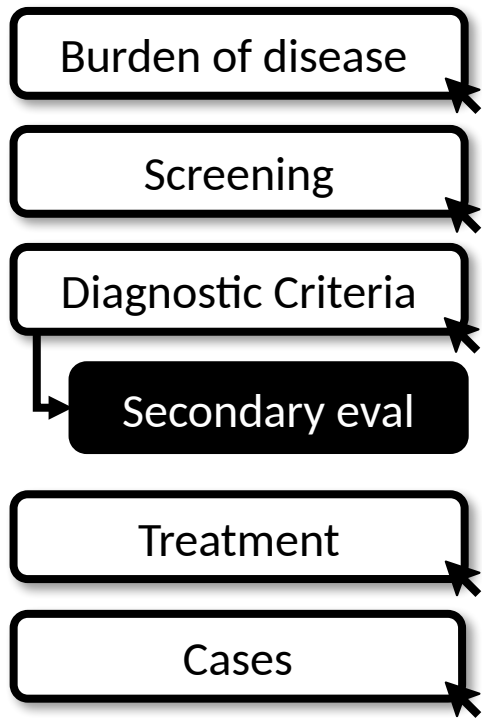
11. Alcohol 3 or more units/day ☒ No ☐ Yes

12. Femoral neck BMD (g/cm²)
Select BMD

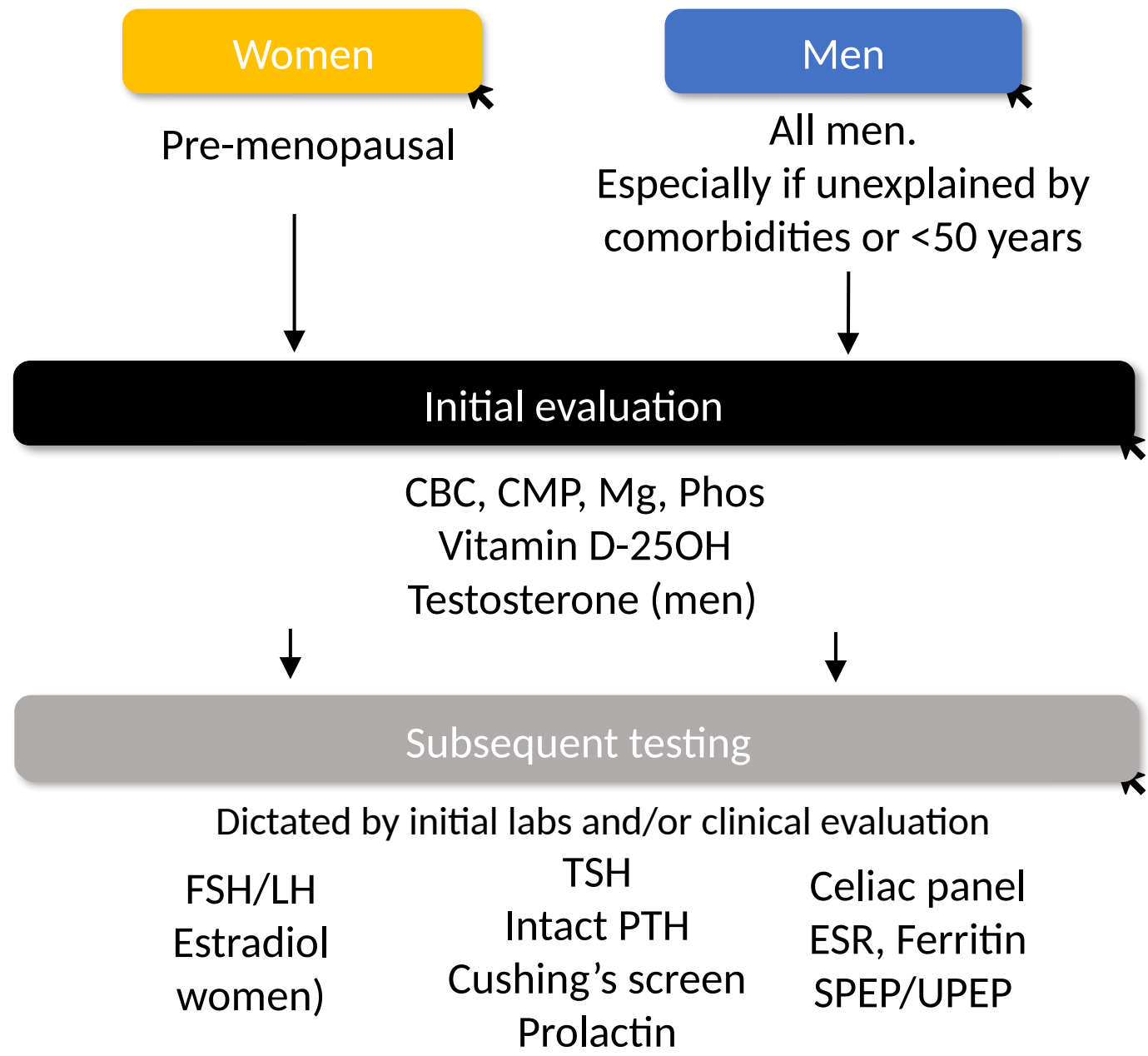
>20% risk of any fracture

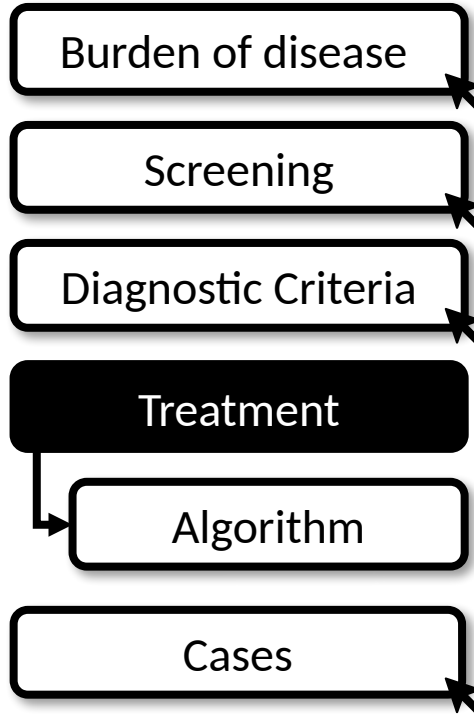
OR

>3% risk hip fracture over 10 years



Of those diagnosed, who needs a secondary work-up?





Burden of disease

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Diagnostic Criteria

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Algorithm

Cases

Primary Interventions



Vitamin D

800 IU daily

(50,000 IU qWeek if Vit D <40)

Ca²⁺

Calcium

1200 mg daily

(ideally from diet)

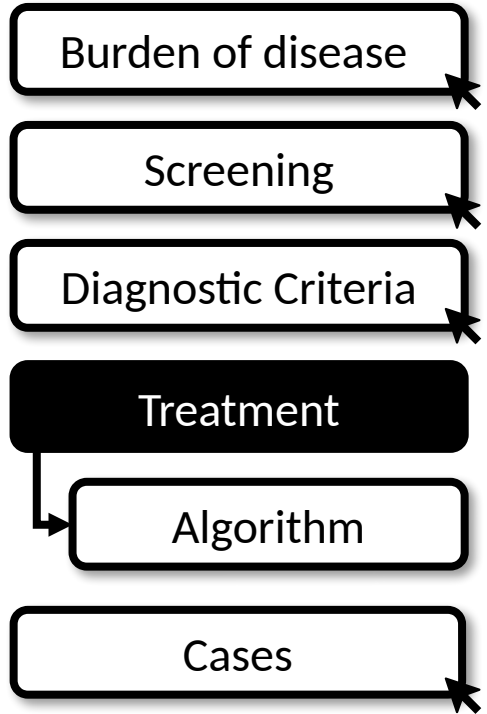
Targeted Therapies



Smoking Cessation



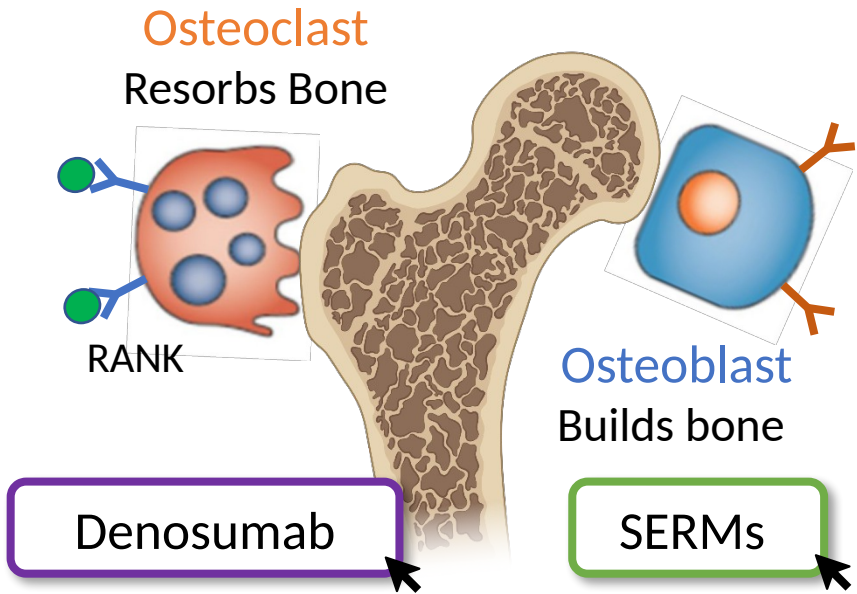
Weight-bearing exercise



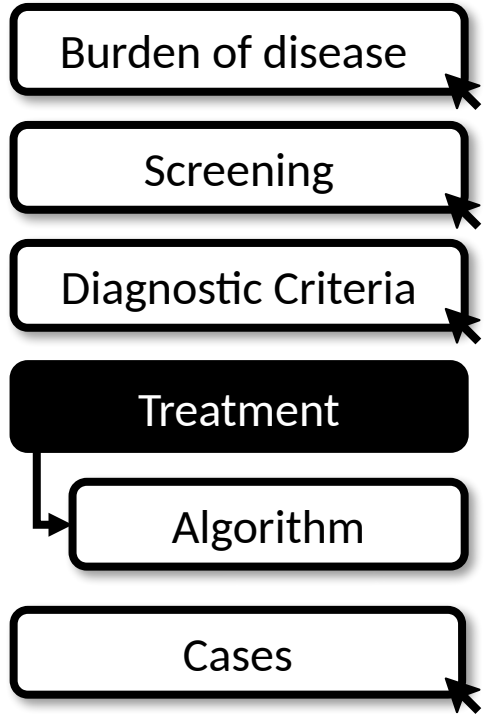
Primary Interventions

Targeted Therapies

Bisphosphonate



Anabolics



Primary Interventions

Targeted Therapies

Bisphosphonate

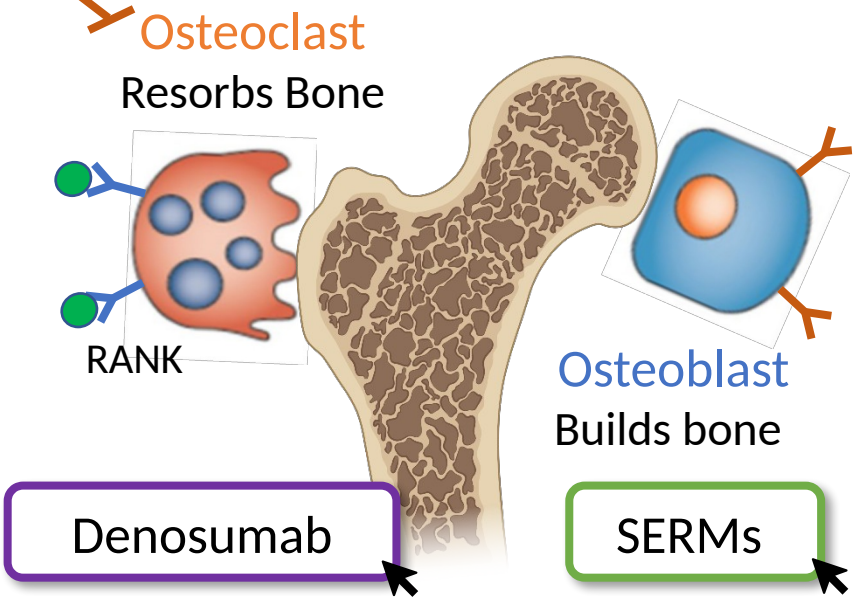
Alendronate (PO weekly)
Zoledronic acid (IV yearly)

1st line for most, GFR >35

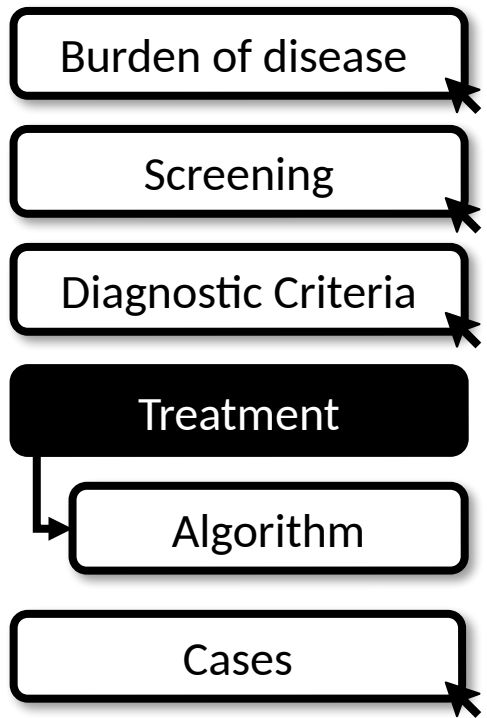
Adverse effects:

- Esophagitis (PO)
- Hypocalcemia
- Atypical femur fracture
- Osteonecrosis of jaw

Consider drug holiday after 5 years (PO) or 3 years (IV)



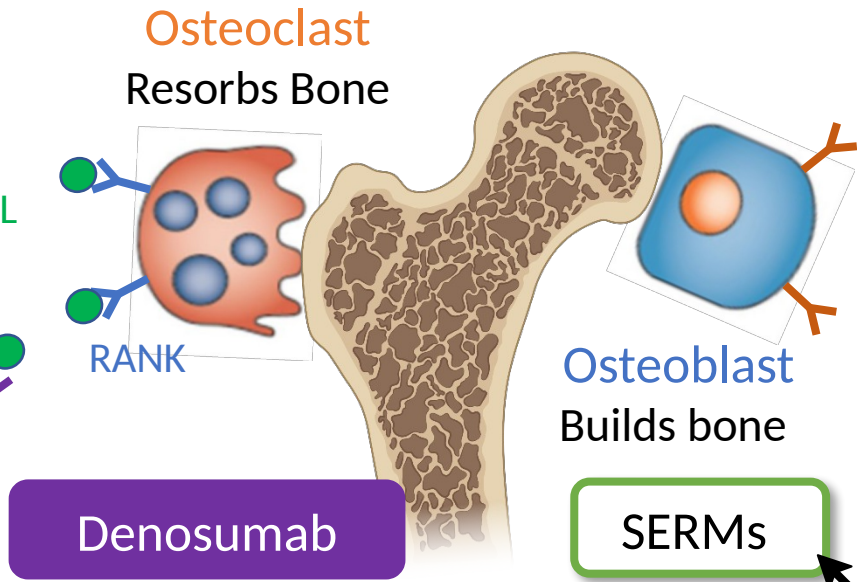
Anabolics



Primary Interventions

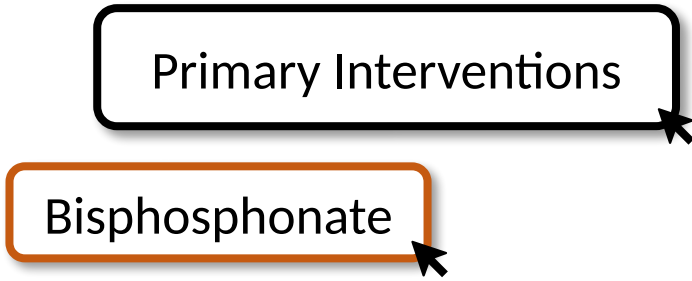
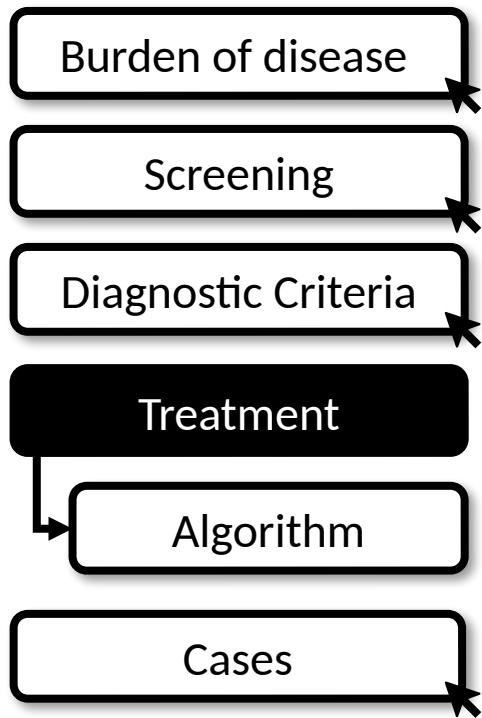
Bisphosphonate

Targeted Therapies



Anabolics

Denosumab/Prolia (SQ Q6 months)
• 2nd line, lifetime therapy • Intolerant or refractory to bisphosphonate • GFR <35
• Rapid bone loss and increased risk of vertebral fractures after discontinuation

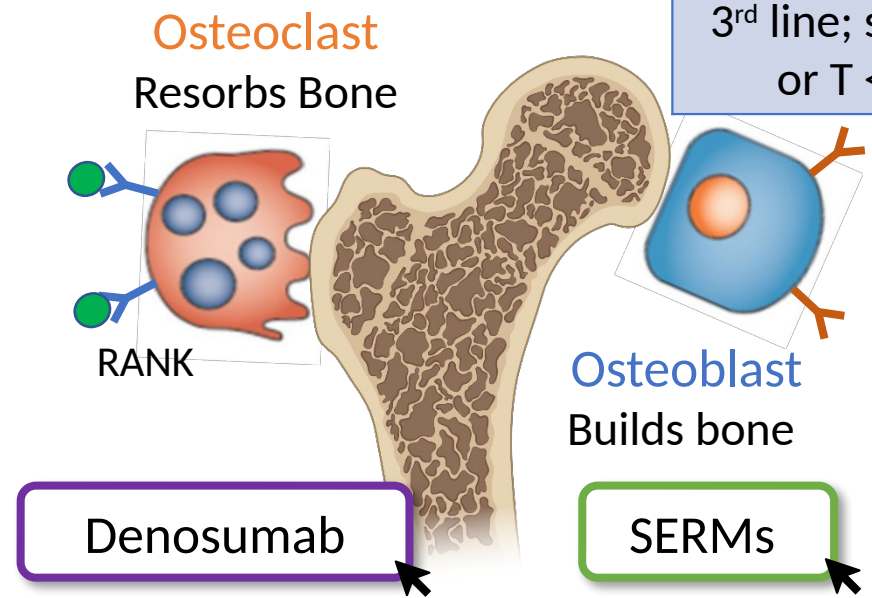


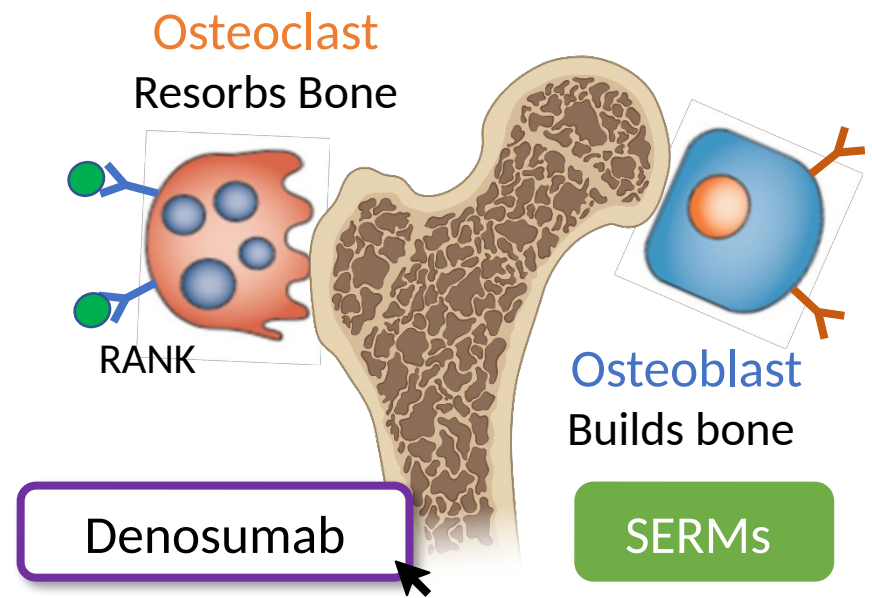
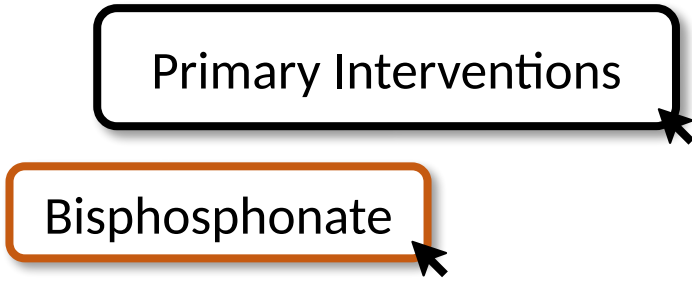
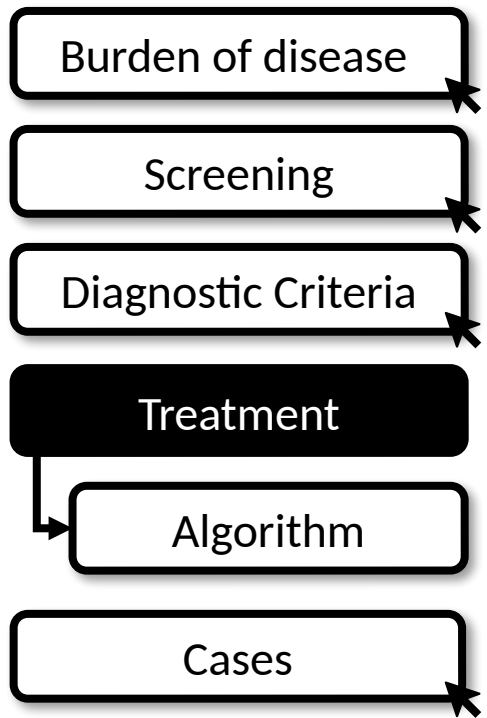
Targeted Therapies

Romozosumab (SQ monthly)
Teriparatide (SQ daily)

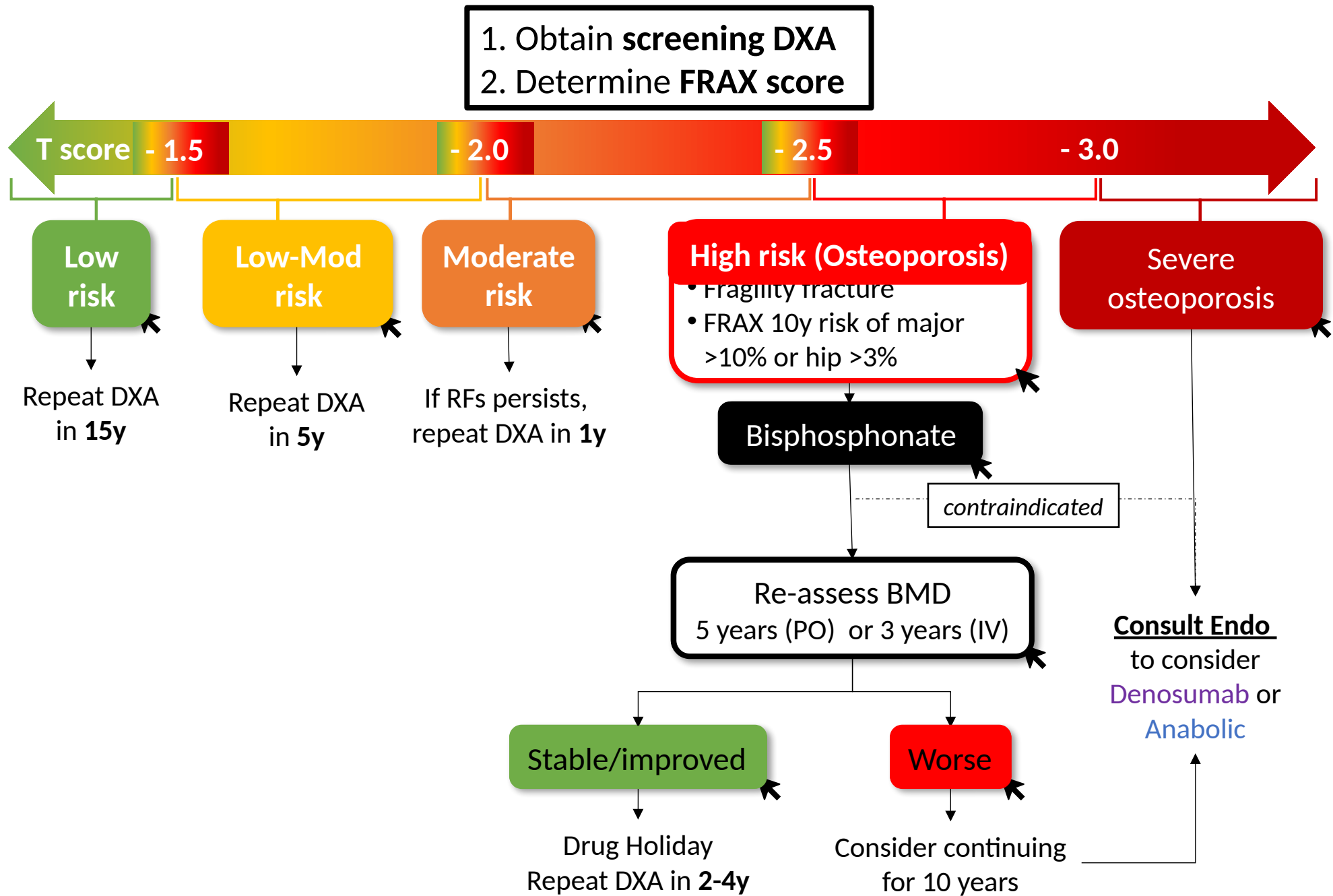
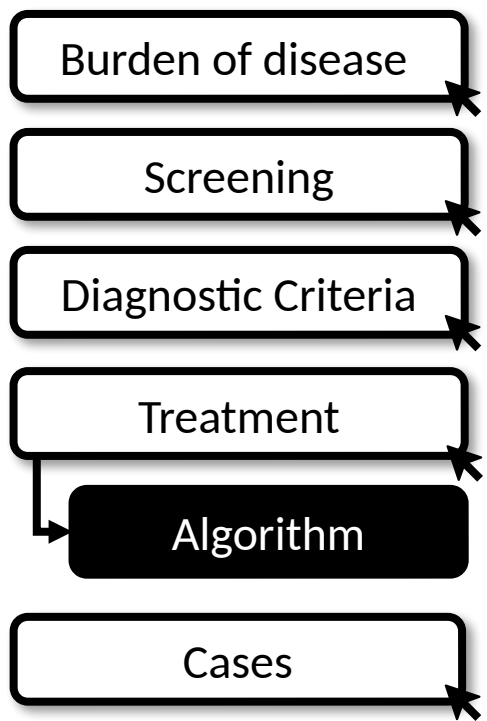
3rd line; severe osteoporosis ($T < -3$)
or $T < -2.5$ + fragility fracture

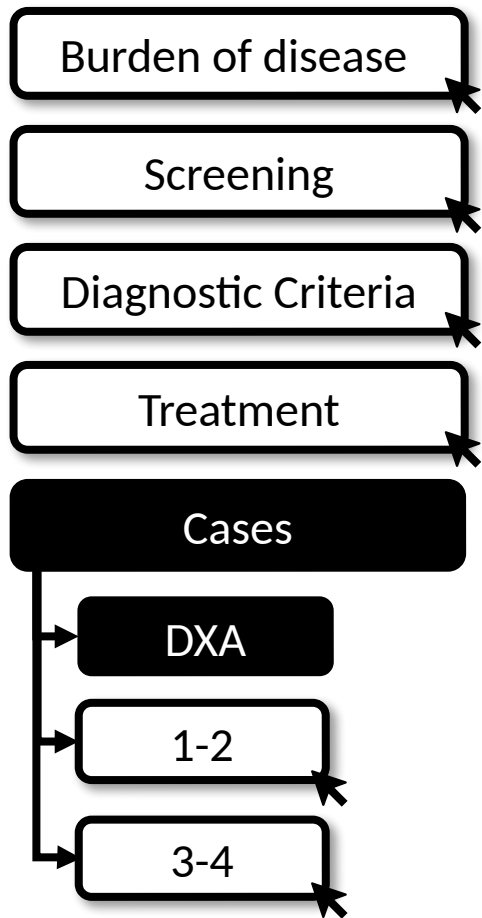
Anabolics





Raloxifene/Evista (PO daily)
• Rarely used, typically 5 year duration • Selective estrogen receptor modifier • Consider in post-menopausal women who can't take bisphosphonate/ denosumab and/or high risk for breast cancer
<i>Adverse effects:</i> ↑ risk of DVT, hot flashes





For the following clinical situations, when would you consider repeating a DXA scan?

65 year-old with normal DXA (low risk)

15 years

60 year-old with osteopenia $T = -2.2$ (moderate risk)

1-2 years.

Osteoporosis on bisphosphonate (IV)

3 years

Osteoporosis on bisphosphonate (PO)

5 years

During bisphosphonate holiday

2-4 years

Burden of disease

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Treatment

Cases

DXA

1-2

3-4

Case 1

A 72 year-old man is evaluated for a 2-week history of low back pain. He has a history of alcohol and tobacco use disorder (50 pack years) and stage 3 chronic kidney disease.

Weight 70kg, 160cm tall. A lumbar XR shows a compression fracture of L4. A DXA scan shows a T-score of -2.2 in the lumbosacral spine and -1.7 in the left hip.

What is his FRAX score?

10-year risk of: major fracture = 15%; hip fracture = 6.5%

What is the diagnosis?

Osteoporosis, based on fragility fracture and FRAX score.
(Note DXA T-score of -1.7 is only osteopenic range)

Is a secondary work-up indicated?

No. Osteoporosis likely explained by age and identified risk factors (tobacco use, alcohol use).

What is the next step in management?

- Encourage smoking cessation, ↓ alcohol intake and ↑ weight-bearing physical activity
- Begin vitamin D and calcium supplementation.
- Offer oral bisphosphonate

Case 2

A 64 year-old woman with a history of hypertension, rheumatoid arthritis, and type II DM is admitted to the hospital after slipping on ice and sustaining a left femoral neck fracture.

What diagnostic workup should be done?

Check vitamin D, BMP, calcium, DXA

T-score at femoral neck is -2.7.
What treatment should you start?

Calcium, vitamin D, weight bearing exercise, and oral bisphosphonate

What are some side effects of bisphosphonates?

- Esophagitis - take with full glass of water and stand upright >30min to reduce risk.
- Osteonecrosis of jaw and atypical fractures are very rare

How long do you continue bisphosphonate therapy?

Re-assess BMD in 5 years with DXA scan. If stable, can consider a drug holiday.

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Cases

DXA

1-2

3-4

Case 3

A 65 year-old-man with presents to clinic with acute atraumatic lower back pain and fatigue. He has no known chronic medical conditions and has never smoked. Lumbar XR shows a L1 compression fracture.

What additional workup is warranted?

He has osteoporosis based on a fragility fracture. Initial work-up should include **CBC, CMP, Vit D, Mg, Phos, testosterone** (in men) and a **DXA scan**. TSH and PTH are often checked early.

...His results show the following:

- WBC 6.3, Hgb 13.1
- **Cr 2.4, Ca 10**, albumin 2.7, total protein 10.8, ALP 200
- TSH: 1.08 | PTH: 27 (15-63) | Vit D: 41
- Testosterone: 438 (230-800)
- T score at femoral neck = -2.7

What additional work-up would you order?

Initial evaluation is unrevealing. Based on his age, hypercalcemia and renal insufficiency, **this is likely multiple myeloma (MM)**. **SPEP/UPEP** would be next step.

He is diagnosed with MM and is started on treatment. What additional therapy is indicated?

Vitamin D and **bisphosphonate** are first line therapy. Likely would **avoid calcium** with active MM.

Case 4

A 74year-old woman with severe COPD and osteoporosis who presents to clinic following an ED visit for an atraumatic vertebral compression fracture following a coughing spell. Her T-score was -2.8 at femoral neck 5 years ago and she has been on alendronate since. Her repeat DXA reveals T-score of -3.1 at the femoral neck.

What is the next step in management?

Anabolic agents should be considered when T-score is < -3.0 OR T-score is <-2.5 with a vertebral fracture.

Options include: teriparatide or romosozumab

When would you repeat a DXA?

2 years

...Her repeat DXA 2 years later is stable with a T-score of -3.0.

What is the next step in management?

Resume bisphosphonate or denosumab

Practice Update — ACP 2023 & ASBMR/BHOF 2024

- **The deck's framework is current and reinforced by the 2023 ACP living guideline (Ann Intern Med). Key points to surface:**
- Bisphosphonates remain FIRST-LINE: strong recommendation in postmenopausal women, conditional in men with primary osteoporosis.
- Denosumab is SECOND-LINE — for bisphosphonate intolerance/contraindication (e.g., eGFR <35).
- VERY-HIGH-RISK women (e.g., $T \leq -3.5$, or $T \leq -2.5$ + fragility fracture, recent/imminent fracture): use an anabolic FIRST (romosozumab or teriparatide) THEN follow with a bisphosphonate. Sequence matters — anabolic-first then antiresorptive gives the greater BMD/fracture benefit.
- Never stop denosumab without transitioning to a bisphosphonate — abrupt cessation causes rebound vertebral fractures. Do NOT follow denosumab with a PTH analog.
- Avoid romosozumab if recent MI/stroke (CV signal). Treat-to-target / goal-directed therapy is the emerging frame (ASBMR/BHOF 2024).

Sources: ACP, Ann Intern Med 2023 (M22-1034); ASBMR/BHOF goal-directed position statement, JBMR 2024.

Take-Home Points

- Screen all women ≥ 65 (and postmenopausal women < 65 or men with major risk factors) with DXA. Diagnose osteoporosis by T-score ≤ -2.5 , a fragility fracture, or a high FRAX score.
- Counsel everyone on calcium, vitamin D, weight-bearing exercise, and tobacco/alcohol reduction. Most fractures occur in people with T-scores better than -2.5 — use clinical risk, not BMD alone.
- Bisphosphonates are first-line for most (eGFR ≥ 35). Counsel on rare harms: esophagitis, osteonecrosis of the jaw, atypical femur fracture. Denosumab second-line; anabolics for very-high-risk.
- Consider a drug holiday after 5 yr oral / 3 yr IV bisphosphonate; re-screen during the holiday every 2-4 yr.

Attribution & Optimization Notes

- Adapted from TeachIM.org "Management of Osteoporosis" (Saxon, deQuillfeldt, Jankousky, Guinn, Wagoner; eds. Brett, Fainstad, Zhang). Published Feb 2022, last reviewed Mar 2023.
- teachim.org/teaching_material/management-of-osteoporosis/ — Licensed CC BY-NC 4.0 (attribution, non-commercial). Educational use only; clinician-reviewed.
- Surgical correction to original deck: "Anabolic steroids" → "Anabolic agents" (Case 3, slide 19) — teriparatide/romosozumab are anabolic agents, not anabolic steroids.
- Added (source-cited): ACP 2023 living-guideline update slide (first-/second-line sequencing, anabolic-first for very-high-risk, denosumab transition rule).
- Human review: slide 16 algorithm lists FRAX "major >10%" for the high-risk/treatment node, while the diagnostic slide (7) and take-home use ">20% major or >3% hip". Left unchanged pending clinician confirmation of intended threshold.
- Companion files in package: Facilitator Guide 2023 (DOCX), Learner Handout — Treatment Options (PPTX), Learner Handout — Treatment Pathway (PDF). Optimization run 2026-06-10.